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HARVARD UNIVERSITY

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Academic Appointments

Harvard University (2023 – Present)
Postdoctoral Fellow, Environment and Natural Resources Program, Harvard Kennedy School Belfer Center & Salata Institute for Climate and Sustainability

Carnegie Mellon University (2022 – 2023)
Postdoctoral Associate, Department of Engineering and Public Policy
Visiting Researcher, Institute of Energy, Environment, and Economy, Tsinghua University (Summer 2023)

Education

Massachusetts Institute of Technology, Institute for Data, Systems, and Society
Ph.D. in Social and Engineering Systems, 2023 (defended September 2022)
- Dissertation: *Local official and polluter accountability in China's environmental inspections*

M.Fin., MIT Sloan School of Management (2017)

Zhejiang University
B.S. in Applied Mathematics (2011–2015)

Fields

Energy and Environmental Policy; Regulation and Policy Implementation
Methods: equilibrium modeling; causal inference; text-as-data methods

Publications

Policy Interactions and the Cost of Decarbonizing Aviation (first author, with Kristen McCormack, William A. Scott, Aaron Smith, Jingran Zhang, and James H. Stock) [link] [code]
Journal of the Association of Environmental and Resource Economists, forthcoming. NBER Working Paper No. 34326.
Contribution: designed the equilibrium model, carried out the calibration, coding, and empirical analysis, and wrote the manuscript.
Policy impact: analysis of California's proposed SAF tax credit with this model was cited by the California Legislative Analyst's Office; covered by E&E News.

Dynamic Responses of SO_2 Pollution to China's Environmental Inspections (co-first author with Valerie J. Karplus) [link] [code]
Proceedings of the National Academy of Sciences, 2023

Working Papers

Expectations and the Cost of Overlapping Climate Policy: Evidence from U.S. Transportation
Develops a dynamic equilibrium model of U.S. fuel, vehicle, and aviation markets linked by federal and state credit programs, with credit banking, biofuel plant investment, and electric-vehicle and charger adoption. Estimates how LCFS and RIN holders form expectations and how biofuel capacity responds to expected margins; a learning rule fits the 2016–25 credit-price history where perfect foresight fails. Compares current policy and federal and state reform packages with an equal-abatement carbon price, including welfare and incidence.

Research in Progress

Do State Low Carbon Fuel Standards Complement Federal Biofuel Mandates? (first author, with Kristen McCormack, William A. Scott, Aaron Smith, Jingran Zhang, and James H. Stock)
Extends the aviation model to multiple states with endogenous electric-vehicle adoption and charging investment, and measures what state clean-fuel standards add to federal biofuel mandates in emissions and cost and how a newly adopting state should design one. *To be presented at the APPAM Fall Research Conference, 2026.*

When Inspectors Return: Who Receives Substantive Enforcement?
Uses case records for 80,000+ citizen complaints handled across two rounds of China's central environmental inspections. Preliminary results: once a complaint is substantiated, the rate of escalated enforcement against small service businesses falls by about 36% relative to heavy industry between rounds, while substantiation rates do not change, placing the shift in local officials' discretion over sanctions. *Presented at the CU Environmental & Resource Economics Workshop, 2026.*

Policy Writing

High Costs and Few Benefits from California's Proposed Sustainable Aviation Fuel Tax Credit (with Aaron Smith, Kristen McCormack, William A. Scott, James H. Stock, and Jingran Zhang) [link]
Energy Institute at Haas Blog, February 2026; cited by the California Legislative Analyst's Office.

Towards Credible and Cost-Effective Options for Sourcing Green Power for Supply Chains in China (with Da Zhang and Valerie J. Karplus) [link]
Policy brief for the Tsinghua Round Table on Corporate Carbon Neutrality Strategies, 2023.

Fellowships

Salata Institute for Climate and Sustainability Postdoctoral Fellowship, Harvard University, 2023–2027
MIT Presidential Fellowship, 2017

Teaching & Advising

Research supervision, Harvard University (2026 – Present)
Supervise undergraduate research assistants on the clean-fuel policy modeling project.

Research supervision, Carnegie Mellon University (2022 – 2023)
Supervised master's research assistants on the Tsinghua green-power supply chain project (interview design, data collection, analysis).

Science, Technology and Public Policy, MIT (Spring 2020, Spring 2021)
Teaching Assistant for Professors Noelle Selin and Kenneth Oye; led weekly recitation sections (10–15 students), grading, and research-project mentoring.

The Economic Approach to Cities and Environmental Sustainability, MIT (Fall 2021)
Guest lecture: "Political Economy of Regulating Pollution"

MIT Kaufman Teaching Certificate Program, MIT (Fall 2020)
Designed course curriculum for "Data Science for Decision Making in Public Policy".

Academic Presentations

2027	ASSA Annual Meeting, poster (scheduled)
2026	APPAM Fall Research Conference (scheduled); Tsinghua University; CU Environmental & Resource Economics Workshop; University of Hong Kong; Hong Kong University of Science and Technology; ARCS Research Conference; Renmin University of China; China and Energy Fellows Seminar, Belfer Center; Fudan University; Huazhong University of Science and Technology; Institutional and Organizational Economics Academy; CES NA Annual Conference
2024	ASSA Annual Meeting
2023	ARCS Annual Research Conference
2022	Northeast Workshop on Energy Policy and Environmental Economics

	2020–21	Environmental Politics and Governance Conference; Duck Family Graduate Workshop in Environmental Politics and Governance (2020, 2021); Institutional and Organizational Economics Academy; ETH Academy on Sustainability and Technology; China Research Group Doctoral Symposium; Interdisciplinary Ph.D. Workshop in Sustainable Development
Policy & Industry Presentations	2026	North American SAF Conference and Expo; International Fuel Ethanol Workshop & Expo
	2025	Regional SAF Workshop for Massachusetts and New England
	2024	North American SAF Conference and Expo
	2023	Roundtable on Corporate Carbon Neutrality Strategies, Tsinghua University

Service

Referee: *GCB Bioenergy*
Student organizer, Sloan China Seminar Series, MIT (2017–2020)
Research Collaborator, Sustainable PPE Initiative, MIT (2020–2021)

Languages

English (fluent), Chinese (native)

Software skills

Python, GAMS, Stata, R, MATLAB, LaTeX

References

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